Redes

Sección 10

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Laboratorio 3 – Algoritmos de Enrutamiento

PDF 2: Evidencia de configuración

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Salidas reales tomadas de Packet Tracer sobre el archivo *Lab3_Algoritmos_Enrutamiento.pkt*, después de configurar OSPF, EIGRP y la redistribución mutua.

1. Tablas de enrutamiento (show ip route) de los 5 routers

Leyenda de códigos (igual para todos los routers):

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

R-OSPF-1

```
R-OSPF-1#show ip route
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.1/32 is directly connected, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/1
L       10.0.0.5/32 is directly connected, Serial0/0/1
O       10.0.0.8/30 [110/20] via 10.0.0.6, 00:00:16, Serial0/0/1
     172.16.0.0/24 is subnetted, 1 subnets
O E2    172.16.0.0/24 [110/20] via 10.0.0.6, 00:00:16, Serial0/0/1
O    192.168.0.0/24 [110/21] via 10.0.0.2, 00:00:16, Serial0/0/0
                    [110/21] via 10.0.0.6, 00:00:16, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30
L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.30
```

R-OSPF-2

```
R-OSPF-2#show ip route
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.2/32 is directly connected, Serial0/0/0
O       10.0.0.4/30 [110/20] via 10.0.0.10, 00:00:31, Serial0/0/1
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.9/32 is directly connected, Serial0/0/1
     172.16.0.0/24 is subnetted, 1 subnets
O E2    172.16.0.0/24 [110/20] via 10.0.0.10, 00:00:31, Serial0/0/1
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/21] via 10.0.0.1, 00:00:21, Serial0/0/0
                       [110/21] via 10.0.0.10, 00:00:21, Serial0/0/1
O       192.168.1.64/28 [110/21] via 10.0.0.1, 00:00:21, Serial0/0/0
                        [110/21] via 10.0.0.10, 00:00:21, Serial0/0/1
```

R-CENTRAL

```
R-CENTRAL#show ip route
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
O       10.0.0.0/30 [110/30] via 10.0.0.9, 00:00:26, Serial0/0/1
                    [110/30] via 10.0.0.5, 00:00:26, Serial0/0/0
C       10.0.0.4/30 is directly connected, Serial0/0/0
L       10.0.0.6/32 is directly connected, Serial0/0/0
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.10/32 is directly connected, Serial0/0/1
D       10.0.0.12/30 [90/2425856] via 10.0.0.21, 00:00:38, Serial0/1/1
                     [90/2425856] via 10.0.0.17, 00:00:37, Serial0/1/0
C       10.0.0.16/30 is directly connected, Serial0/1/0
L       10.0.0.18/32 is directly connected, Serial0/1/0
C       10.0.0.20/30 is directly connected, Serial0/1/1
L       10.0.0.22/32 is directly connected, Serial0/1/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 4 masks
S       172.16.0.0/24 is directly connected, Null0
D       172.16.0.0/25 [90/1914112] via 10.0.0.21, 00:01:20, Serial0/1/1
D       172.16.0.128/26 [90/1916416] via 10.0.0.17, 00:01:19, Serial0/1/0
D       172.16.0.192/27 [90/1916416] via 10.0.0.17, 00:01:19, Serial0/1/0
D    192.168.0.0/23 is a summary, 00:01:08, Null0
O    192.168.0.0/24 [110/11] via 10.0.0.9, 00:01:08, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/11] via 10.0.0.5, 00:01:08, Serial0/0/0
O       192.168.1.64/28 [110/11] via 10.0.0.5, 00:01:08, Serial0/0/0
```

R-EIGRP-1

```
R-EIGRP-1#show ip route
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/6766080] via 10.0.0.18, 00:00:26, Serial0/0/1
D EX    10.0.0.4/30 [170/6766080] via 10.0.0.18, 00:00:36, Serial0/0/1
D EX    10.0.0.8/30 [170/6766080] via 10.0.0.18, 00:00:37, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/0
L       10.0.0.13/32 is directly connected, Serial0/0/0
C       10.0.0.16/30 is directly connected, Serial0/0/1
L       10.0.0.17/32 is directly connected, Serial0/0/1
D       10.0.0.20/30 [90/6510080] via 10.0.0.18, 00:00:36, Serial0/0/1
                     [90/6766080] via 10.0.0.14, 00:00:41, Serial0/0/0
     172.16.0.0/16 is variably subnetted, 5 subnets, 4 masks
D       172.16.0.0/25 [90/2170112] via 10.0.0.14, 00:00:41, Serial0/0/0
                      [90/6510336] via 10.0.0.18, 00:00:36, Serial0/0/1
C       172.16.0.128/26 is directly connected, GigabitEthernet0/0.10
L       172.16.0.129/32 is directly connected, GigabitEthernet0/0.10
C       172.16.0.192/27 is directly connected, GigabitEthernet0/0.20
L       172.16.0.193/32 is directly connected, GigabitEthernet0/0.20
D    192.168.0.0/23 [90/6766080] via 10.0.0.18, 00:00:26, Serial0/0/1
```

R-EIGRP-2

```
R-EIGRP-2#show ip route
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/6766080] via 10.0.0.22, 00:00:26, Serial0/0/1
D EX    10.0.0.4/30 [170/6766080] via 10.0.0.22, 00:00:37, Serial0/0/1
D EX    10.0.0.8/30 [170/6766080] via 10.0.0.22, 00:00:37, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/0
L       10.0.0.14/32 is directly connected, Serial0/0/0
D       10.0.0.16/30 [90/6510080] via 10.0.0.22, 00:00:37, Serial0/0/1
                     [90/6766080] via 10.0.0.13, 00:00:41, Serial0/0/0
C       10.0.0.20/30 is directly connected, Serial0/0/1
L       10.0.0.21/32 is directly connected, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 4 masks
C       172.16.0.0/25 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
D       172.16.0.128/26 [90/2172416] via 10.0.0.13, 00:00:41, Serial0/0/0
                        [90/6512640] via 10.0.0.22, 00:00:37, Serial0/0/1
D       172.16.0.192/27 [90/2172416] via 10.0.0.13, 00:00:41, Serial0/0/0
                        [90/6512640] via 10.0.0.22, 00:00:37, Serial0/0/1
D    192.168.0.0/23 [90/6766080] via 10.0.0.22, 00:00:26, Serial0/0/1
```

Notas: en R-OSPF-1 y R-OSPF-2 la red 172.16.0.0/24 aparece como **O E2** (resumen del lado EIGRP). En R-EIGRP-1 y R-EIGRP-2 las redes de OSPF aparecen como **D EX** (enlaces) y como una sola ruta **D 192.168.0.0/23** (resumen del lado OSPF). R-CENTRAL genera ambos resúmenes: **S 172.16.0.0/24 -> Null0** y **D 192.168.0.0/23 is a summary**.

2. Balanceo de costo igual en OSPF

Entre R-OSPF-1 y R-OSPF-2 existen dos caminos: el enlace directo (costo 20) y el camino por R-CENTRAL (10 + 10 = 20). Las interfaces seriales de R-OSPF-1 lo muestran con `show ip ospf interface`:

```
R-OSPF-1#show ip ospf interface Serial0/0/0
Serial0/0/0 is up, line protocol is up
  Internet address is 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 20
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:07
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)

R-OSPF-1#show ip ospf interface Serial0/0/1
Serial0/0/1 is up, line protocol is up
  Internet address is 10.0.0.5/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 10
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:07
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 3.3.3.3
  Suppress hello for 0 neighbor(s)
```

Resultado: R-OSPF-1 instala **dos rutas de igual costo [110/21]** hacia la misma red 192.168.0.0/24 (LAN de R-OSPF-2), una por cada serial:

```
R-OSPF-1#show ip route
...
O    192.168.0.0/24 [110/21] via 10.0.0.2, 00:00:16, Serial0/0/0
                    [110/21] via 10.0.0.6, 00:00:16, Serial0/0/1
```

R-OSPF-2 también ve dos rutas de costo 21 hacia las VLANs de R-OSPF-1 (192.168.1.0/26 y 192.168.1.64/28), según la tabla de la sección 1. Vecindades OSPF de R-OSPF-1:

```
R-OSPF-1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           0   FULL/  -        00:00:33    10.0.0.6        Serial0/0/1
2.2.2.2           0   FULL/  -        00:00:33    10.0.0.2        Serial0/0/0
```

3. Balanceo de costo desigual en EIGRP (variance 3)

Entre R-EIGRP-1 y R-EIGRP-2 el enlace directo es la ruta principal y la que pasa por R-CENTRAL es la secundaria. Con `variance 3` EIGRP usa ambas en proporción inversa a su métrica.

```
R-EIGRP-1#show ip eigrp topology
IP-EIGRP Topology Table for AS 100/ID(4.4.4.4)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/30, 1 successors, FD is 6766080
         via Rstatic (6766080/2169856)
P 10.0.0.4/30, 1 successors, FD is 6766080
         via Rstatic (6766080/2169856)
P 10.0.0.8/30, 1 successors, FD is 6766080
         via Rstatic (6766080/2169856)
P 10.0.0.12/30, 1 successors, FD is 2169856
         via Connected, Serial0/0/0
P 10.0.0.16/30, 1 successors, FD is 6254080
         via Connected, Serial0/0/1
```

```
P 10.0.0.20/30, 2 successors, FD is 6510080
         via 10.0.0.18 (6510080/1913856), Serial0/0/1
         via 10.0.0.14 (6766080/6254080), Serial0/0/0
P 172.16.0.0/25, 2 successors, FD is 2170112
         via 10.0.0.14 (2170112/2816), Serial0/0/0
         via 10.0.0.18 (6510336/1914112), Serial0/0/1
P 172.16.0.128/26, 1 successors, FD is 28160
 (la salida de Packet Tracer se pagina con --More--; se continua en el detalle)
```

```
R-EIGRP-1#show ip eigrp topology 172.16.0.0 255.255.255.128
IP-EIGRP (AS 100): Topology entry for 172.16.0.0/25
  State is Passive, Query origin flag is 1, 2 Successor(s), FD is 2170112
  Routing Descriptor Blocks:
  10.0.0.14 (Serial0/0/0), from 10.0.0.14, Send flag is 0x0
      Composite metric is (2170112/2816), Route is Internal
      Vector metric:
        Minimum bandwidth is 1544 Kbit
        Total delay is 20010 microseconds
        Reliability is 255/255
        Load is 1/255
        Minimum MTU is 1500
        Hop count is 1
  10.0.0.18 (Serial0/0/1), from 10.0.0.18, Send flag is 0x0
      Composite metric is (6510336/1914112), Route is Internal
      Vector metric:
        Minimum bandwidth is 1544 Kbit
        Total delay is 189550 microseconds
        Reliability is 255/255
        Load is 1/255
        Minimum MTU is 1500
        Hop count is 2
```

Interpretación (destino 172.16.0.0/25 visto desde R-EIGRP-1):

```
Successor principal : via 10.0.0.14 (Serial0/0/0)  métrica 2 170 112   (AD del vecino: 2816)
Segunda ruta (FS)   : via 10.0.0.18 (Serial0/0/1)  métrica 6 510 336   (AD del vecino: 1 914 112)

Condición de factibilidad: AD 1 914 112  <  FD del successor 2 170 112      -> se cumple (feasible successor)
Condición de variance    : 6 510 336 <= 3 x 2 170 112 = 6 510 336           -> se cumple
Relación de métricas     : 6 510 336 / 2 170 112 = 3,0  ->  reparto 3 : 1 = 75 % / 25 %
```

Al estar habilitado variance, PT muestra ambas rutas bajo '2 successors' y las instala en la tabla de rutas (ver sección 1, R-EIGRP-1 y R-EIGRP-2).

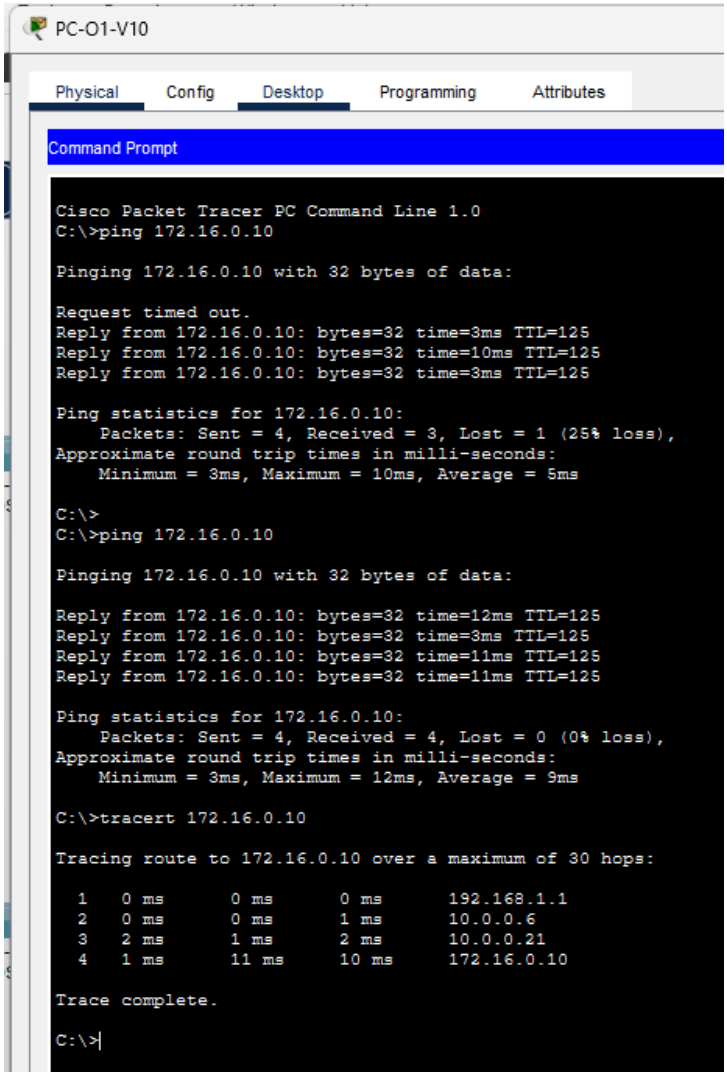
```
R-EIGRP-1#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface      Hold Uptime    SRTT   RTO   Q   Seq
                                   (sec)          (ms)        Cnt  Num
0   10.0.0.14       Se0/0/0        13   00:00:52  40     1000  0   17
1   10.0.0.18       Se0/0/1        13   00:00:48  40     1000  0   51
```

```
R-EIGRP-1#show ip protocols
Routing Protocol is "eigrp  100 "
  Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  Router-ID: 4.4.4.4
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum metric variance 3
  Automatic Summarization: disabled
```

4. Prueba de conectividad extremo a extremo

Desde **PC-O1-V10** (VLAN 10 de R-OSPF-1, 192.168.1.10) hacia **PC-E2** (LAN de R-EIGRP-2, 172.16.0.10), atravesando los dos dominios de enrutamiento.

Captura de pantalla del ping y el tracert exitosos (Command Prompt de PC-O1-V10):



Texto de la misma prueba:

```
C:\>ping 172.16.0.10

Pinging 172.16.0.10 with 32 bytes of data:

Reply from 172.16.0.10: bytes=32 time=12ms TTL=125
Reply from 172.16.0.10: bytes=32 time=3ms TTL=125
Reply from 172.16.0.10: bytes=32 time=11ms TTL=125
Reply from 172.16.0.10: bytes=32 time=11ms TTL=125

Ping statistics for 172.16.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 12ms, Average = 9ms

C:\>tracert 172.16.0.10

Tracing route to 172.16.0.10 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.1.1
  2    0 ms      0 ms      1 ms      10.0.0.6
  3    2 ms      1 ms      2 ms      10.0.0.21
```

```
  4   1 ms      11 ms     10 ms     172.16.0.10

Trace complete.
```

El camino confirma la redistribución: sale por el gateway de la VLAN 10 (192.168.1.1), pasa por R-CENTRAL (10.0.0.6) y entra al dominio EIGRP por R-EIGRP-2 (10.0.0.21). El TTL 125 corresponde a 3 routers atravesados (128 - 3). El primer ping de la captura perdió un paquete mientras se resolvía ARP; al repetirlo hubo 0 % de pérdida.